Auxiliary Variables for Recovering FLS Geographic Sampling Composition

1. Purpose

The entropy-calibration estimator requires two kinds of information:

1. **County features** describing the agricultural operations and workers located in each county.
2. **FLS targets** measuring the corresponding characteristics of the realized survey estimate in a particular region and year.

The strongest auxiliary variables are generated by the same FLS sample and survey weights as the AEWR-setting wage but do not themselves contain the field-and-livestock wage. Employment composition, seasonality, and hours are therefore preferable to alternative FLS wage measures.

The field-and-livestock wage used to set the AEWR should be reserved for validation and first-stage estimation.

2. Priority Summary

Priority	Auxiliary concept	Published FLS target	Preferred county analogue	Recommended role
1	Employment duration	Workers employed 150 days or more versus 149 days or less by region and quarter	Census of Agriculture county workers by days worked	Main calibration target
1	Quarterly seasonality	Regional hired-worker counts in January, April, July, and October	County QCEW NAICS 11 quarterly employment ratios	Main calibration target
1	Weekly hours	Regional gross weekly hours by quarter and annual average	Pooled ACS agricultural-worker hours or predicted county hours	Main or soft target

Priority	Auxiliary concept	Published FLS target	Preferred county analogue	Recommended role
2	Peak worker-size composition	Ideally state/stratum sample and response counts; published national worker-size distribution is insufficient geographically	Census of Agriculture farms and workers by hired-worker count	Prior and NASS data request
2	Farm sales class	Ideally state/stratum sample and response counts	Census of Agriculture farms and employment by sales class	Prior and NASS data request
2	Crop versus livestock composition	Limited national or broad-region FLS tables	Census farm type, cash receipts, CDL, and livestock inventories	Prior, validation, or soft target
3	Occupational composition	National FLS SOC employment totals in recent reports	OEWS or ACS agricultural employment by SOC	Prior and validation
3	Area-frame representation	Ideally list-versus-area sample and response aggregates	Small farms, acreage, pasture, and farm-count proxies	Prior and NASS data request
Avoid	Field-and-livestock wage	Published FLS/AEWR-setting wage	OEWS field-and-livestock wage	Outcome and validation only

3. Employment Duration

FLS variable

For each region and quarterly reference week, the FLS reports:

- workers expected to be employed 150 days or more;
- workers expected to be employed 149 days or less;
- total hired workers.

Construct

$$T_{rtq}^{long} = \frac{N_{rtq}^{150+}}{N_{rtq}^{150+} + N_{rtq}^{149-}}.$$

County analogue

The Census of Agriculture reports hired workers by number of days worked. Construct

$$z_{it}^{long} = \frac{N_{it}^{150+}}{N_{it}^{150+} + N_{it}^{149-}}.$$

Additional useful county variables are:

- number of long-duration workers;
- number of short-duration workers;
- long-duration workers per farm;
- long-duration share of estimated worker hours;
- change in the duration composition between Census years.

Advantages

Employment duration is closely related to seasonality, farm type, and operation size. It also resembles an important dimension of the FLS published employment tables.

Limitations

Census of Agriculture values are observed only every five years. Reasonable alternatives are:

1. use the latest preceding Census value;
2. linearly interpolate between Census years;
3. estimate a county duration-share model using Census years and annual predictors.

All three should be compared in sensitivity analysis.

4. Quarterly Employment Seasonality

FLS variable

The FLS reports hired-worker counts for January, April, July, and October reference weeks. Convert the levels into ratios relative to the annual or quarterly average:

$$T_{rt}^{emp,q} = \frac{N_{rtq}^{FLS}}{\frac{1}{4}\sum_{q'} N_{rtq'}^{FLS}}.$$

Alternatively construct quarterly shares:

$$S_{rt}^{emp,q} = \frac{N_{rtq}^{FLS}}{\sum_{q'} N_{rtq'}^{FLS}}.$$

Only three of the four quarterly shares are independent.

Preferred county analogue

Use QCEW NAICS 11 quarterly employment:

$$z_{it}^{emp,q} = \frac{E_{itq}^{QCEW}}{\frac{1}{4}\sum_{q'} E_{itq'}^{QCEW}}.$$

Potential alternatives are:

- monthly state agricultural employment assigned to counties using county production shares;
- county H-2A seasonal positions;
- crop acreage combined with crop-specific labor calendars;
- county agricultural employment seasonality estimated from ACS or administrative data.

Advantages

Quarterly seasonality can distinguish areas with spring, summer, or fall employment peaks. Random overrepresentation of counties with different seasonal patterns should appear in the FLS quarterly employment profile.

Limitations

County QCEW cells are frequently suppressed. Suppression must be addressed using a documented allocation or imputation model. H-2A employment is related to the policy environment and should be used only as a sensitivity variable.

5. Average Weekly Hours

FLS variable

The FLS publishes gross weekly hours for each region and quarterly reference week:

$$T_{rtq}^{hours} = H_{rtq}^{FLS}.$$

It also publishes an annual average hours measure.

County analogue

Possible county measures include:

1. pooled ACS average weekly hours for hired agricultural employees;
2. predicted county hours from a model estimated on pooled ACS microdata;
3. QCEW hours where available;
4. state hours interacted with county crop mix, seasonality, and worker duration.

The county feature is

$$z_{itq}^{hours} = \widehat{H}_{itq}.$$

Advantages

Hours are directly relevant to the effective contribution of a farm to the FLS wage ratio. A county with more worker hours should receive more effective weight even when its headcount is unchanged.

Limitations

County ACS samples of agricultural workers are small. A prediction model or multiyear pooling is preferable to raw county-year means. Hours should generally be a soft rather than exact calibration target.

6. Peak Worker-Count Composition

Connection to the FLS design

The FLS list-frame sample is stratified principally by peak worker counts or calculated farm sales. Therefore, the distribution of farms and workers across worker-size categories is central to expected sampling intensity.

County variables

Using the Census of Agriculture, construct:

- farms with 1 hired worker;
- farms with 2 workers;
- farms with 3–6 workers;
- farms with 7–10 workers;
- farms with 11–20 workers;
- farms with 21–50 workers;
- farms with 51 or more workers;
- estimated workers in each class;
- average workers per farm;
- share of county workers employed on large-worker farms;
- concentration of hired workers across farm-size classes.

A compact feature vector might be

$$\mathbf{z}_{it}^{size} = \begin{bmatrix} \text{small-farm worker share} \\ \text{medium-farm worker share} \\ \text{large-farm worker share} \end{bmatrix},$$

with one omitted category where necessary.

Published target limitation

Recent FLS reports publish a national distribution of workers by number of workers on the farm. A national distribution cannot reveal which counties within a particular AEWR region were overrepresented. These variables are therefore best used to improve the expected-weight prior unless NASS supplies state- or region-by-stratum sample information.

Recommended role

Use worker-size composition in the prior and in sensitivity analysis. Elevate it to a primary calibration variable if state-by-stratum sample or usable-response counts can be obtained.

7. Farm Sales-Class Composition

Connection to the FLS design

Calculated farm value of sales is used when worker-count information is unavailable or as another dimension of the hierarchical sampling strata.

County variables

From the Census of Agriculture, construct:

- farms by gross sales class;
- share of farms in each sales class;
- share of agricultural sales in each class;
- sales per farm;
- hired workers per sales class, if available;
- hired-labor expense associated with large operations;
- share of county employment associated with farms above selected sales thresholds.

Published target limitation

FLS reports include national or broad-region statistics by economic class, but these do not consistently provide 17-region realized sample shares. They are therefore weak as direct year-by-region calibration targets.

Recommended role

Use sales-class composition to improve the prior. Request state-by-sales-stratum sample and response information from NASS.

8. Crop and Livestock Composition

County variables

Potential sources and measures include:

- Census of Agriculture crop and livestock cash receipts;
- number of crop, livestock, dairy, poultry, nursery, and greenhouse farms;
- hired-labor expenses by farm type;
- CDL acres by crop group;
- pasture and rangeland acres;
- cattle, dairy cow, poultry, hog, and other livestock inventories;
- crop-to-livestock employment ratio;
- specialty-crop share;
- perennial-versus-annual crop share.

A useful reduced feature vector is

$$\mathbf{z}_{it}^{type} = \begin{bmatrix} \text{field-crop share} \\ \text{specialty-crop share} \\ \text{livestock share} \end{bmatrix}.$$

Because the shares sum to one, omit one category in the optimizer.

FLS information

Recent reports contain:

- national field-and-livestock worker shares by type of farm;
- field-and-livestock wages by farm type for broader combined regions;
- separate regional field and livestock wages.

The national and broad-region tables are not ideal targets for annual calibration within all 17 AEWR regions. Separate field and livestock wages are also too closely connected to the AEWR-setting wage to serve as preferred auxiliary variables.

Recommended role

Use crop and livestock composition in the prior, for validation, and potentially as a soft calibration target when geographically comparable FLS employment shares can be recovered.

9. Occupational Composition

County variables

Using OEWS or ACS, estimate employment shares for:

- graders and sorters of agricultural products;
- agricultural equipment operators;
- crop, nursery, and greenhouse workers;
- farm, ranch, and aquacultural-animal workers;
- other agricultural workers;
- hand packers and packagers;
- agricultural supervisors and managers.

For occupation k, define

$$z_{it}^{occ,k} = \frac{E_{it}^{occ,k}}{\sum_{\ell} E_{it}^{occ,\ell}}.$$

FLS information

Recent FLS reports publish national SOC employment and wage tables. National employment totals are useful for checking the OEWS occupational crosswalk but do not identify region-specific geographic tilts.

Recommended role

Use occupational employment shares as prior predictors and validation measures. Do not use SOC wages as main calibration targets.

10. Area-Frame and List-Frame Proxies

The FLS area-frame nonoverlap sample represents agricultural operations not included on the list frame. Potential county proxies include:

- number of small farms;
- farms with low sales;
- farms reporting no hired workers in the Census year;
- agricultural acreage per listed or observed farm;
- cropland, pasture, and rangeland acreage;
- small-farm share of county agricultural acreage;
- farm-count-to-employment ratio;
- number of new or very small operations;
- prevalence of part-time or retirement farms.

These variables should generally alter expected representation rather than serve as exact calibration targets. They become direct calibration variables only if NASS provides list-frame and area-frame sample or response totals by state or region.

11. Response-Propensity Variables

Variables such as rural broadband access, farm operator age, business organization, language, and farm complexity could predict survey response. However, without FLS response counts by geography or stratum, their relationship to realized response adjustment cannot be estimated credibly.

Recommended use:

- include selected variables in a prior-response model only if external evidence supports it;
- use them to test whether inferred tilts track observable response determinants;
- prioritize obtaining actual response counts rather than constructing a highly parameterized response model.

12. Variables to Avoid as Main Calibration Targets

The preferred estimator should not use:

- contemporaneous FLS field-and-livestock wages;
- the AEWR level or change;
- separate contemporaneous field and livestock FLS wages;
- state field-and-livestock FLS wages;
- all-hired FLS wages as a primary target;
- contemporaneous local outcome variables;
- crop prices or weather shocks as target moments;
- H-2A certifications as a major target moment.

The first five contain or are closely connected to the AEWR-setting wage. The latter variables can reflect actual economic shocks rather than sampling composition.

Weather, crop prices, H-2A exposure, and local economic outcomes remain valuable falsification variables. An inferred sampling tilt should not systematically predict them after conditioning on the features used to construct the weights.

13. Recommended Initial Target Set

The initial implementation should remain low dimensional. A reasonable four-moment specification is:

1. average share of workers employed 150 days or more;
2. July employment relative to annual average employment;
3. October employment relative to annual average employment;
4. average weekly hours.

The county analogues would be:

1. Census of Agriculture long-duration worker share;
2. county QCEW July-quarter employment ratio;
3. county QCEW October-quarter employment ratio;
4. pooled-ACS or predicted weekly hours.

A richer specification could add January and April seasonality, crop-versus-livestock composition, and large-farm employment share. These should be added only after checking common support, collinearity, and weight stability.

14. Recommended Prior Features

Even when no matching annual FLS target exists, the following should be considered when constructing or evaluating the prior:

- hired agricultural employment;
- estimated worker hours;
- worker-size composition;
- farm sales-class composition;
- long- versus short-duration worker shares;
- crop, specialty-crop, and livestock shares;
- agricultural seasonality;
- farm count and small-farm prevalence;
- list-frame completeness proxies;
- state and AEWR-region membership.

The simplest defensible prior remains employment or employment-hours. More elaborate priors should be presented as sensitivity specifications unless their coefficients can be derived from documented FLS sampling rates.

15. Highest-Value NASS Data Request

The most useful additional information would be nondisclosive aggregates by:

- year;
- April versus October survey wave;
- state;
- list versus area frame;
- peak-worker or sales sampling stratum.

For each cell, request:

1. number of farms on the frame;
2. number sampled;
3. number eligible;
4. number responding;
5. number providing usable employment reports;
6. number providing usable wage and hours reports;
7. initial sampling fraction;
8. total or mean initial expansion weight;
9. total or mean final nonresponse-adjusted weight;
10. expanded worker count;
11. expanded worker hours.

If only a subset can be released, prioritize usable-response counts and final weight totals by state, peak-worker stratum, and survey wave. These aggregates are closest to the realized sampling randomness of interest.

16. Data-Source Map

Concept	Candidate source	Geography	Frequency	Main limitation
Expected hired employment	QCEW NAICS 11	County where disclosed	Quarterly/annual	Suppression
Farm employment	BEA	County	Annual	May include proprietors or differ from FLS scope
Worker duration	Census of Agriculture	County	Every five years	Interpolation required
Farms by worker size	Census of Agriculture	County	Every five years	May not reproduce exact FLS peak-worker strata

Concept	Candidate source	Geography	Frequency	Main limitation
Farms by sales class	Census of Agriculture	County	Every five years	Sales classes may require harmonization
Crop acreage	CDL	County	Annual	Acreage is not employment
Farm type and receipts	Census of Agriculture	County	Every five years	Interpolation required
Livestock inventories	Census of Agriculture/NASS	County	Annual or five-year, depending on series	Suppression and uneven coverage
Agricultural occupation mix	OEWS	OEWS area/county allocation	Annual	Not genuinely county-specific in many areas
Worker hours	ACS	County with pooling	Annual/multiyear	Small samples
FLS duration, employment, and hours targets	FLS reports	AEWR region	Quarterly/annual	Sampling error and concept matching
FLS sample and response counts	NASS request	Ideally state by stratum	Survey wave	Not routinely published

17. Official Methodology Sources

- USDA National Agricultural Statistics Service, Agricultural (Farm) Labor Survey.
- USDA National Agricultural Statistics Service, Farm Labor Methodology and Quality Measures archive.
- USDA National Agricultural Statistics Service, Farm Labor Methodology and Quality Measures, February 2021.